Name ______________________________ Section ________

MTH:2310 DISCRETE MATH
QUIZ 8
DUE: MONDAY APRIL 14

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: *(4 points)*
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 2 points for arithmetic errors.
You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. *(12 points) Basic Modular Arithmetic*
Express the following integers in the form $a = qd + r$ as in the Division Algorithm.

(a) *(3 points)* $a = 419$, $d = 4$

$419 = 104 \cdot 4 + 3$

(b) *(3 points)* $a = -200$, $d = 6$

$-200 = -34 \cdot 6 + 4$

(c) *(3 points)* $a = -48$, $d = 7$

$-48 = -7 \cdot 7 + 1$

(d) *(3 points)* $a = 555$, $d = 5$

$555 = 111 \cdot 5 + 0$

Problem 2. *(12 points) Basic Modular Arithmetic*
Determine whether the following statements are true or false. **Explain why.**

(a) *(6 points)* $419 \equiv 371 (\text{mod } 4)$

$419 \ (\text{mod } 4) = 3$

$371 \ (\text{mod } 4) = 3$

so $419 \equiv 371 \ (\text{mod } 4)$

(b) *(6 points)* $-9999 \equiv 3344 (\text{mod } 101)$

$-9999 \ (\text{mod } 101) = 0$

$3344 \ (\text{mod } 101) = 11$

so $-9999 \not\equiv 3344 \ (\text{mod } 11)$

Problem 3. *(10 points) Basic Modular Arithmetic*
Evaluate these quantities.

(a) *(4 points)* $(67^{100} \cdot 5^2 + 35 \cdot 16)(\text{mod } 33)$

$= \left[\left((67 \text{ mod } 33)^{100} \text{ mod } 33 \right) \cdot \left(5 \text{ mod } 33 \right)^2 \text{ mod } 33 + (35 \text{ mod } 33)(16 \text{ mod } 33) \text{ mod } 33 \right] \text{ mod } 33$

$= \left[(1^{100} \text{ mod } 33) \cdot (25 \text{ mod } 33) \text{ mod } 33 + (2 \cdot 16) \text{ mod } 33 \right] \text{ mod } 33$

$= [25 + 32] \text{ mod } 33 = 57 \text{ mod } 33 = \boxed{24}$

(b) *(6 points)* $\Big((201 \cdot 3 \cdot 14)(\text{mod } 5) - (42 \cdot 15)(\text{mod } 8) \Big)(\text{mod } 3)$

$= \left[(201 \text{ mod } 5)(3 \text{ mod } 5)(14 \text{ mod } 5) \text{ mod } 5 - (42 \text{ mod } 8)(15 \text{ mod } 8) \text{ mod } 8 \right] \text{ mod } 3$

$= [(1 \cdot 3 \cdot 4) \text{ mod } 5 - (2 \cdot 7) \text{ mod } 8] \text{ mod } 3 = [12 \text{ mod } 5 - 14 \text{ mod } 8] \text{ mod } 3$

$= [2 - 6] \text{ mod } 3 = -4 \text{ mod } 3 = \boxed{2}$

Problem 4. *(8 points) Basic Modular Arithmetic*
Find the multiplicative inverse of 4 modulo 7 if it exists. If not, explain why.

If 4^{-1} mod 7 exists, then

$4 \cdot 4^{-1} \equiv 1 \ (\text{mod } 7)$

4^{-1} could be 0, 1, 2, 3, 4, 5, or 6

Notice

$4 \cdot (2) \text{ mod } 7 = 8 \text{ mod } 7 = 1$

So $\boxed{4^{-1} \text{ mod } 7 = 2}$

Name ______________________________ Section ________

Problem 5. *(8 points) Basic Modular Arithmetic*
Find the multiplicative inverse of 12 modulo 20 if it exists. If not, explain why.

If $12^{-1} \bmod 20$ exists, then

$12 \cdot 12^{-1} \equiv 1 \pmod{20}$

12^{-1} could be 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, or 19

For any integer N, if $N \equiv 1 \pmod{20}$

then $N = 1 + 20 \cdot k$ for an integer k.

so $N = 2 \cdot (10k) + 1$. Then N must be odd.

On the other hand, for any integer M,

$12 \cdot M = 2 \cdot (6 \cdot M)$ is always even.

Since $12 \cdot x$ is always even, there are no solutions to $12 \cdot 12^{-1} \equiv 1 \pmod{20}$.

Problem 6. *(12 points) Primes and Greatest Common Divisors*
Use Trial Division to determine whether the following integers are prime.

(a) *(6 points)* 201

$\sqrt{201} \approx 14$

Must check primes 2, 3, 5, 7, 11, 13

$2 \mid 201$? NO 201 is odd

$3 \mid 201$? Yes $\frac{201}{3} = 67$

Therefore 201 is not prime.

(b) *(6 points)* 107

$\sqrt{107} \approx 10$

Must check primes 2, 3, 5, 7

$2 \mid 107$? NO 107 is odd

$3 \mid 107$? NO $107 = 35 \cdot 3 + 2$

$5 \mid 107$? NO $107 = 21 \cdot 5 + 2$

$7 \mid 107$? NO $107 = 15 \cdot 7 + 2$

So by Trial Division, 107 is prime.

Problem 7. *(16 points) Primes and Greatest Common Divisors*
Let $a = 2970$, $b = 1260$.

(a) *(4 points)* Find the prime factorization of 2970 and 1260.

$2970 = 2 \cdot 3^3 \cdot 5 \cdot 11$

$1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7$

(b) *(4 points)* What are the least common multiple and greatest common divisor of 2970 and 1260 using their prime factorizations?

$\mathrm{lcm}(1260, 2970) = 2^2 \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 = 41{,}580$

$\gcd(1260, 2970) = 2 \cdot 3^2 \cdot 5 = 90$

(c) *(8 points)* Find the greatest common divisor of 2970 and 1260 using the Euclidean algorithm. **Show every step**.

$r_1 = 2970 \bmod 1260 = 450$

$r_2 = 1260 \bmod 450 = 360$

$r_3 = 450 \bmod 360 = 90$

$r_4 = 360 \bmod \boxed{90} = \underline{0}$

So $\gcd(2970, 1260) = 90$